

Your grade: 100%

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Next item →

1.

In CART, what does the Gini index measure?

1 / 1 point

☒ The impurity or class distribution in a node

☐ How spread out the target values are around the mean in a node

☐ The median distance between nodes

☒ Correct

Correct! The Gini index measures the impurity or class distribution in a node.
2.

Which of the following are cons with using decision trees (select all that apply)?

1 / 1 point

☒ They are inefficient with linear relationships

☒ Correct

Correct! Trees don't deal well with linear relationships (they create step functions, which are inefficient).

☒ They can be unstable

☒ Correct

Correct! Decision trees can be unstable. Small changes in the training dataset can create a completely different tree. Why? Because each split depends on the parent split. If a different feature is selected as the first split feature, the entire tree structure will be changed.

☐ You can use features without any transformations
3.

In Decision Rules, what is the default rule?

1 / 1 point

☐ The rule that applies before other rules

☒ The rule that applies when no other rule applies

☐ The rule that applies before training occurs

☒ Correct

Correct! The default rule is the rule that applies when no other rule applies. The prediction of the default rule is typically the most frequent class of the data points which are not covered by other rules.
4.

Which of the following is NOT an algorithm for learning rules?

1 / 1 point

☐ RIPPER

☐ SLIPPER

☒ DIPPER

☒ Correct

Correct. Dipper is not an algorithm for learning decision rules.
5.

Fill in the blank: RuleFit learns a sparse linear model with the _____ and also a number of new features that are decision rules.

1 / 1 point

☒ original features

☐ default features

☐ raw features

☒ Correct

Correct! RuleFit learns a sparse linear model with the original features and also a number of new features that are decision rules.
6.

What is the combination of rule learning algorithm and linear model used in RuleFitRegressor in the imodels python library?

1 / 1 point

☐ The imodels library uses a tree ensemble generated with gradient boosting to learn rules and a Ridge linear regression model.

☒ The imodels library uses a tree ensemble generated with gradient boosting to learn rules and a Lasso linear regression model.

☐ The imodels library uses a tree ensemble generated without gradient boosting to learn rules and a linear regression model.

☒ Correct

Correct! The imodels library uses a tree ensemble generated with gradient boosting to learn rules and a Lasso linear regression model.
7.

What are advantages to using ProtoPNet (select all that apply)?

1 / 1 point

☒ Interpretable by design

☒ Correct

Correct! ProtoPNet is interpretable by design (not post-hoc).

☐ Additional training complexity

☒ Comparable accuracy to baseline models

☒ Correct

Correct! In the original paper, it showed comparable accuracy to baseline models and combined models reached SOTA accuracy.
8.

MonoNet consists of two blocks. What are the two blocks (check all that apply)?

1 / 1 point

☒ Monotonic Block

☒ Correct

Correct! The Monotonic Block in MonoNet enforces a monotonic relationship between hidden layer and output.

☒ Unconstrained Block

☒ Correct

Correct! The unconstrained block in MonoNet learns arbitrary representations.

☐ Constrained Block
9.

What does Kolmogorov Arnold Networks NOT have that makes them revolutionary (and interpretable by design)?

1 / 1 point

☒ Linear weights

☐ Activation functions

☐ Sparsity

☒ Correct

Correct! KANs have no linear weights at all – every weight parameter is replaced by a univariate function parametrized as a spline
10.

What type of spline is used by KANs?

1 / 1 point

☒ B splines

☐ C splines

☐ Y splines

☒ Correct

Correct! Every weight parameter in a KAN is replaced by a univariate function parametrized as a B spline.